# How To Install And Confi n Ubuntu

By **SK**

## About QEMI

As a Linux enthu

QEMU is a generic and open source machin zer. It is a free and open source tool that allows users to create achines inside the host operating system. The resources of t em, such as Hard drive, RAM, Processor, will be divided and operating systems (Virtual machines).

When used as a machine emulator, QEMU ams made for one machine (e.g. an ARM board) on a differe own PC). By using dynamic translation, it achieves very good p

When used as a virtualizer, QEMU achieve nces by executing the guest code directly on the host CPU. QE on when executing under the Xen hypervisor or using the KVM kernel m in Linux. When using KVM, QEMU can virtualize x86, server and embedded PowerPC, and S390 guests.

**QEMU has two operating modes:**

- Full system emulation: In this mode, QEMU emulates a full system (for example a PC), including a processor and various peripherals. It can be used to launch different Operating Systems without rebooting the PC or to debug system code.
- User mode emulation (Linux host only): In this mode, QEMU can launch Linux processes compiled for one CPU on another CPU. For example, it can be used to launch Wine or to ease cross-compilation and cross-debugging.

QEMU can run without a host kernel driver and yet gives acceptable performance.

**QEMU will work fine in the following hardware:**

- Mac99 PowerMac (PowerPC processor, in
- Sun4m/Sun4c/Sun4d (32-bit Sparc proces
- Sun4u/Sun4v (64-bit Sparc processor, in
- Malta board (32-bit and 64-bit MIPS proce
- MIPS Magnum (64-bit MIPS processor) ;
- ARM Integrator/CP (ARM) ;
- ARM Versatile baseboard (ARM) ;
- ARM RealView Emulation/Platform basebo
- Spitz, Akita, Borzoi, Terrier and Tosa PDAs
- Luminary Micro LM3S811EVB (ARM Corte
- Luminary Micro LM3S6965EVB (ARM Cort
- Freescale MCF5208EVB (ColdFire V2) ;
- Arnewsh MCF5206 evaluation board (ColdFire V2) ;
- Palm Tungsten|E PDA (OMAP310 processor) ;
- N800 and N810 tablets (OMAP2420 processor) ;
- MusicPal (MV88W8618 ARM processor) ;
- Gumstix “Connex” and “Verdex” motherboards (PXA255/270) ;
- Siemens SX1 smartphone (OMAP310 processor) ;
- AXIS-Devboard88 (CRISv32 ETRAX-FS) ;
- Petalogix Spartan 3aDSP1800 MMU ref design (MicroBlaze) ;
- Avnet LX60/LX110/LX200 boards (Xtensa).

For user emulation, x86 (32 and 64 bit), PowerPC (32 and 64 bit), ARM, MIPS (32 bit only), Sparc (32 and 64 bit), Alpha, ColdFire(m68k), CRISv32 and MicroBlaze CPUs are supported.

## 1. Install QE

In this let us see h
create and manag

## 2. Creating Virtual Machine

Now, let us create a new virtual machine. Fc :orial, let us create a new Ubuntu 15.04 server.

First create the Virtual hard disk image foı ıple, let us create 20GB size hard disk image.

```
qemu-img create ubuntu.img 20G
```

Or you can create the image with Qemu's ıat 'qcow2' using the following command:

```
qemu-img create -f qcow2 ubuntu.qcow 20G
```

Then, download the Ubuntu 15.04 server installation image and boot the Virtual machine. I already downloaded the installation image and kept it in my home directory.

To boot the Virtual machine, run:

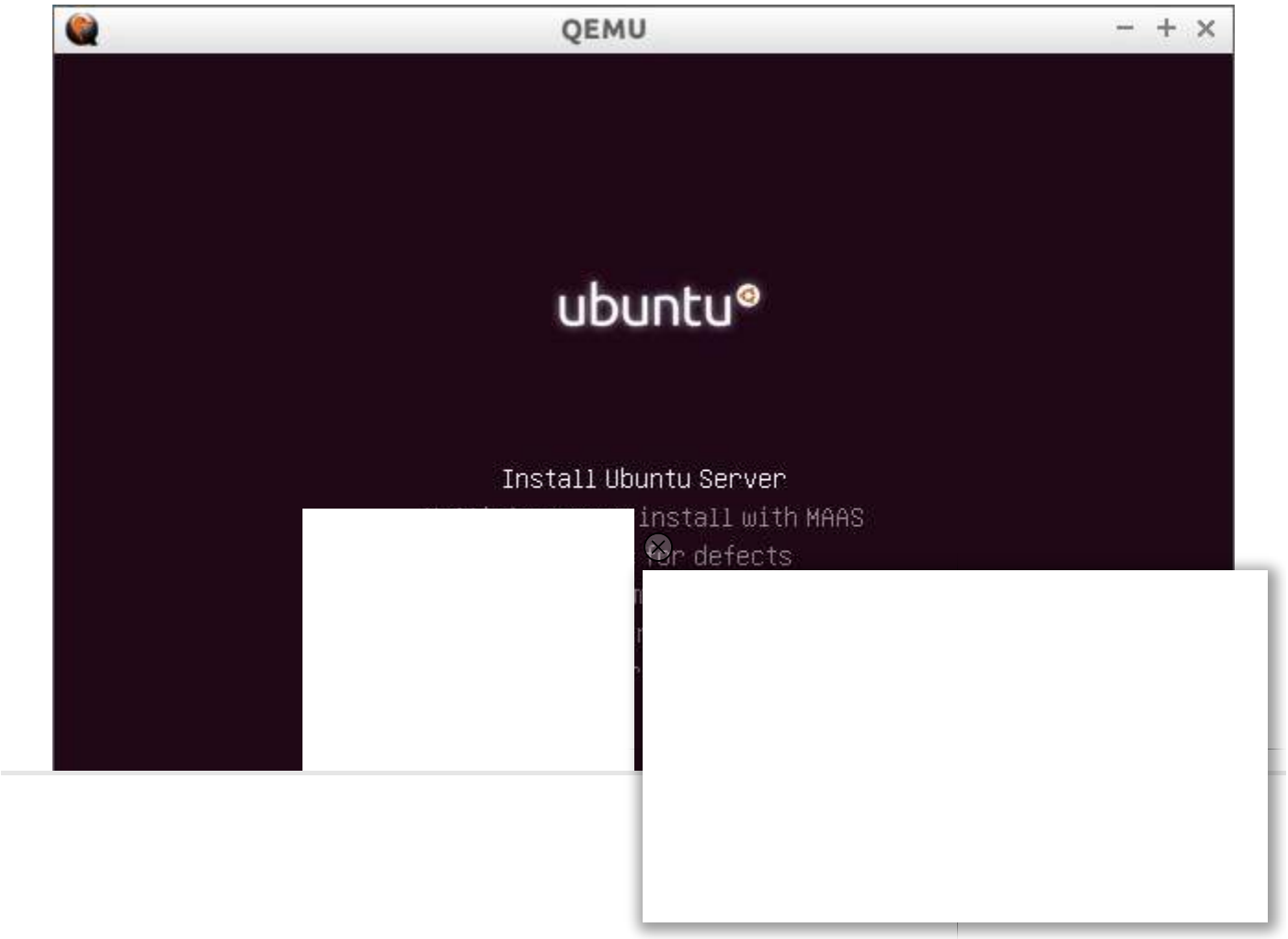
```
qemu-system-x86_64 -hda ubuntu.img -boot d -cdrom /home/sk
/Soft_Backup/OS\ Images/New/ubuntu-15.04-server-amd64.iso -m 640
```

For 32bit VM, use:

```
qemu-system-xi:  -boot d -cdrom /home/sk
/Soft_Backup/O:  -
```

Here, I have alloca u

Беларуская
Български
Bengali
Tibetan
Bosanski
Català
Čeština
Dansk
Deutsch
Dzongkha
Ελληνικά
English
Esperanto
Español
Eesti
Euskara
Suomi
Gujarati
Hindi
Hrvatski
Magyar
Bahasa Indonesia
Íslenska
Italiano
日本語
Kazak
Khmer
한국어
Kurdî
Lao
Lietuviškai
Latviski
Tagalog
Türkçe
Uyghur
Українська
Tiếng Việt
中文(简体)
中文(繁體)
Svenska
F1 Help F2 Language F3 Keymap F4 Modes F5 Accessibility F6 Other Options


QEMU
ubuntu
Install Ubuntu Server
install with MAAS

```
u -hda ubuntu.img -m 640
```

That's it.

Don't like command line way? Creating an the command line might be bit difficult to some user. No worri /Ms using GUI tool called Virtual Machine Manager (Virt Manag

## 3. Create and manage Vi using Virt Manager

Launch **Virtual Machine Manager**(Virt n he Menu or Unity Dash. The default Interface of Virt Manager

reate a virtual machine, go to **File -> N**

Select how would like to install the operatin

You can install guest OS using local ISO ima tall, PXE server, or importing existing disk images. Also, you c architecture from the Advanced options.

Select the local ins

Choose the amount of RAM and no of CPUs.

Enter your virtual h ult is 8GB.

Enter your guest OS name. And, Select the Advanced options. Finally, click Finish.

Once you click the

That’s it. Now, continue and install the Gues in normal system.

To view the number of installed guest OSs, go to **File -> View manager**.

To view the Virtual machine details, Selec nd go to **Edit -> Virtual Machine Detail**s.

Click on the ‘Show virtual hardware details’ from the menu bar.

That’s it. Cheers!!

**Reference:**

- **QEMU Website**

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